ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 8 Technology paper 1 · practice b.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 A ✓ 1.2 B ✓ 1.3 D ✓ 1.4 A ✓ 1.5 C ✓ 1.6 C ✓ 1.7 A ✓ 1.8 D ✓ 1.9 B ✓ 1.10 D ✓ [10]

QUESTION 2

2.1.1 Tie ✓ (1)

2.1.2 A triangle is a rigid shape that cannot change shape unless one side breaks or bends ✓, so the truss stays strong and does not collapse under the roof load ✓. (2)

2.2 (a) Any ONE: electricity pylon, roof truss, bicycle frame, scaffolding ✓ (b) Any ONE: car body, helmet, egg shell, cool-drink can ✓ (c) Any ONE: dam wall, brick wall, pyramid ✓ (3)

2.3 The load presses down on the arch, which is in compression ✓; the curved shape spreads the force outwards and down to the supports (abutments) at each end ✓. (2)

2.4 A static load does not move or change, e.g. the weight of the bridge itself ✓; a dynamic load moves or changes, e.g. traffic, people or wind ✓. (2)

2.5 (a) The folds make the sheet deeper so it resists bending ✓ (b) A tube is light but resists bending and twisting almost as well as a solid rod ✓ (c) A gusset covers the joint and makes it rigid, so it does not bend or pull apart ✓ (3)

2.6 Specifications, any TWO: must span a gap of e.g. 40 cm ✓; must hold a mass of e.g. 1 kg ✓; must be wide enough for a toy car; must look attractive. Constraint, any ONE: only straws, string and tape may be used; must be finished in two weeks; limited number of straws ✓ (3)

2.7 To test whether the design is strong and safe and find weaknesses ✓ before spending money and time on the real bridge ✓. (2)

[18]

QUESTION 3

3.1 (a) Any TWO: light ✓; waterproof ✓; cheap; easy to mould into shapes; durable; good electrical insulator (4)
(b) Any ONE explained: plastic does not rot (is not biodegradable), so it fills landfills and pollutes rivers and oceans ✓; animals swallow it or get trapped and die ✓; burning plastic releases toxic gases

3.2 (a) PET (polyethylene terephthalate) ✓ (b) HDPE (high-density polyethylene) ✓ (2)

3.3 A material made by combining two or more different materials to get better properties ✓. Any ONE: fibreglass; reinforced concrete; plywood; carbon fibre ✓ (2)

3.4 A thermoplastic sheet is heated until it is soft ✓ → the soft sheet is placed over a mould ✓ → the air is sucked out so the plastic is pulled tightly over the mould; it cools, hardens and is removed ✓ (3)

3.5 Plastic is an electrical insulator and gives a good, comfortable grip ✓ (1)

[12]

QUESTION 4

4.1 (a) $36 \div 12 = 3:1$ ✓ (b) $1\,500 \times {}^{12}/_{36}$ ✓ $= 500$ r/min ✓ (c) Bigger ✓ (4)

4.2 (a) $1\,400 \times {}^{50}/_{350}$ ✓ $= 200$ r/min ✓ (b) $7:1$ ✓ (3)

4.3 20×5 mm ✓ $= 100$ mm ✓ (2)

4.4 Any TWO: the chain does not slip ✓; it can transmit more power (force) ✓; it lasts longer; it works in wet conditions (2)

4.5 Input: electric motor (switched on by a remote control) ✓ → Process: the motor turns the pinion, which moves along the rack ✓ → Output: the gate slides open or closed ✓ (3)

4.6 It changes the direction of the force ✓: you pull down on the rope to lift the flag up, which is easier and more convenient ✓. (2)

4.7 To reduce friction ✓, which reduces wear and heat and makes the mechanism run smoothly and quietly ✓. (2)

[18]

QUESTION 5

5.1 Correct symbols for battery ✓, lamp ✓ and switches ✓; the two switches connected in parallel with each other and in series with the lamp ✓ (4)

Driver's door	Passenger's door	Light
0	0	0
0	1	1
1	0	1
1	1	1

5.2 Four rows with all input combinations ✓; light column correct ✓✓✓ (one mark for every correct row after the first) (4)

5.3 (a) It limits (reduces) the current ✓ (b) The lamp becomes dimmer ✓ (2)

5.4 Any TWO: washing machine ✓; fridge ✓; fan; blender; vacuum cleaner; hairdryer; electric drill (2)

5.5 A wire connects the metal casing to the ground (earth) ✓; if a fault makes the casing live, the current flows safely to earth and the fuse or breaker trips, so the user does not get a shock ✓. (2)

5.6 Current takes a path with almost no resistance (e.g. bare wires touching) ✓; a very large current flows, which can overheat wires and cause fires ✓. (2)

5.7 OR logic ✓ (1)

[17]

TOTAL: 75

END OF MARKING GUIDELINE